Univariate Time-Series Modelling

The Box–Jenkins Methodology

Identification, Estimation, Diagnostic Checking and Forecasting

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Expanded lecture notes with a conceptual foundation in the theory of oscillations,
intuition, worked examples and a trace of the root literature

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CHAPTER 1

THE OSCILLATOR HERITAGE

1.1 WHY BEGIN WITH OSCILLATORS?

Every model in these notes (AR, MA, ARMA, ARIMA) describes a system whose state today carries the memory of its state yesterday, and which is repeatedly knocked about by random shocks. That is exactly the problem the *theory of oscillations* posed and solved for mechanical and electrical systems, long before time-series econometrics existed. The standard text of that school, Andronov et al. (1966) (first Russian edition 1937), opens with a methodological argument that applies to economists as much as to physicists. Every theoretical study must idealise the real system; how far one may idealise cannot be settled in advance, because it depends on the questions being asked; and the legitimacy of an idealisation is judged either against experiment or by comparing two theories, one with the simplification and one without (Andronov et al., 1966, pp. xv–xvii).

Box–Jenkins is that philosophy turned into a procedure. Identification chooses the idealisation, diagnostic checking confronts it with the data, and forecasting confronts it with the future. This chapter does three things. First, it shows that an AR(2) process is a discretised, noise-driven damped oscillator, so the stability and cycle properties of ARMA models are the familiar phenomena of mechanics. Second, it translates the oscillation theorists' key ideas (stability, coarseness, parasitic parameters) into time-series language. Third, it marks the boundary of the linear framework and traces the two intellectual lineages involved. Beyond the Introduction, the later sections draw on the book's treatment of first-order systems, bifurcation theory, the method of slowly varying amplitudes, point transformations and a model from mathematical biology. Two further sources are used: Pikovsky et al. (2001) on self-sustained oscillators and their synchronisation, and Ginoux (2015) on how Poincaré's limit cycles reached the theory of oscillations.

> IN PLAIN WORDS. Pluck a guitar string and it rings at one pitch while the sound fades. Now imagine rain falling continuously on the string: it never stops sounding, the pitch is still the string's own, but the loudness wanders unpredictably. The first picture is a damped

oscillator; the second is the same oscillator driven by random shocks, which is what an autoregression describes.

1.2 THE HARMONIC OSCILLATOR AND ITS PHASE PLANE

The simplest oscillator, a mass on springs without friction, obeys

$$\ddot{x} + \omega_0^2 x = 0, \qquad x(t) = K\cos(\omega_0 t + \alpha), \tag{1.1}$$

where the frequency ω_0 belongs to the system, while the amplitude K and phase α are fixed by the initial conditions (Andronov et al., 1966, pp. 1–3). The key device of the school is the *phase plane*: put $y = \dot{x}$ and treat the pair (x, y) as the *state* of the system. Each state is a point; each motion is a curve; the frictionless oscillator fills the plane with nested ellipses around a single equilibrium, which they call a *centre* (pp. 4–5, 8). Rewriting the second-order equation as two first-order equations gives $dy/dx = -\omega_0^2 x/y$, from which the shape of the curves can be read off *without solving the equation*. The authors call this *qualitative integration*: knowing the portrait tells us the character of every possible motion, which is what matters when closed-form solutions are unavailable (pp. xxxi, 6–10).

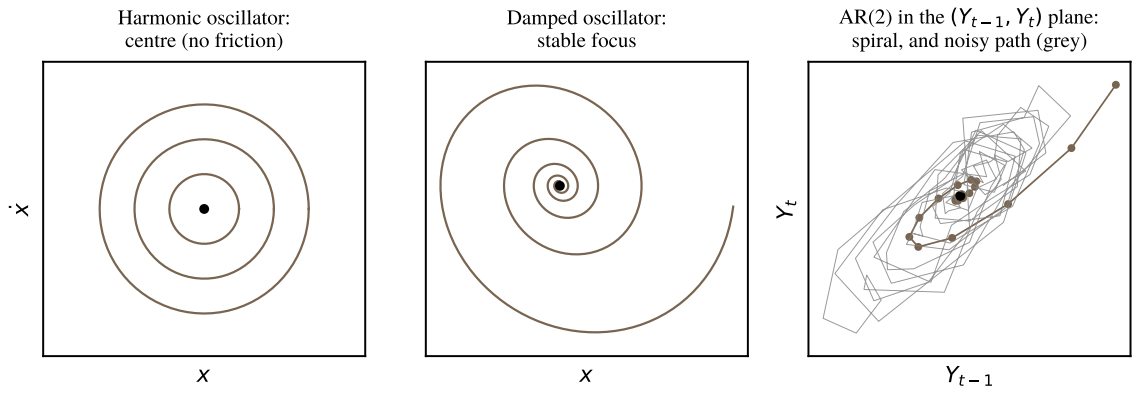


Figure 1.1: Left: phase portrait of the frictionless harmonic oscillator (a centre). Middle: with friction the paths spiral into the equilibrium (a stable focus). Right: the AR(2) with $\phi_1 = 1.34$, $\phi_2 = -0.65$ drawn in the plane of (Y_{t-1}, Y_t): the deterministic part spirals inwards just like the damped oscillator; adding shocks (grey) keeps the point circulating around the equilibrium.

The time-series translation is immediate. To predict Y_{t+1} from an AR(2) one needs the pair (Y_t, Y_{t-1}), which is the discrete-time state vector. Writing the process as a two-dimensional

first-order system with companion matrix

$$\begin{pmatrix} Y_t \\ Y_{t-1} \end{pmatrix} = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} Y_{t-1} \\ Y_{t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ 0 \end{pmatrix}$$

makes the analogy exact. In the oscillation theorists' counting, the number of degrees of freedom is *half* the number of state variables (p. xix), so an AR(2) is a one-degree-of-freedom oscillator, an AR(1) is what they call a system with "half a degree of freedom", and an AR(p) has $p/2$.

1.3 HALF A DEGREE OF FREEDOM: THE LEAKY BUCKET AND THE AR(1)

When friction is strong and the inertia (mass or inductance) is negligible, the second-order equation collapses to $b\dot{x} + kx = 0$. The authors call this a system with *half a degree of freedom*: one number fixes the state, the phase space is a line rather than a plane, and every motion is the aperiodic decay $x(t) = x_0 e^{-kt/b}$; an *RC* circuit is the electrical example (Andronov et al., 1966, Chapter I, §5, from p. 36). The same section makes a point that carries over exactly: in this idealisation the initial velocity can no longer be assigned independently of x_0, because the equation itself fixes it. An AR(1) has the same property: Y_{t-1} is the entire state, and there is no separate "velocity" to specify.

Observe the system every Δ time units and the free motion obeys $Y_t = \phi Y_{t-1}$ with

$$\phi = e^{-\Delta/\tau}, \qquad \tau = b/k \ (= RC \text{ for the circuit}). \tag{1.2}$$

If the system is also driven by white noise, the continuous-time process is the Ornstein–Uhlenbeck process, $dY = -Y\,dt/\tau + \sigma\,dW$, and its sampled values are *exactly* an AR(1) with this ϕ and innovation variance $\frac{1}{2}\sigma^2\tau(1-\phi^2)$ (Uhlenbeck and Ornstein, 1930; Doob, 1942). This is unlike the oscillator, whose sampled version is only an ARMA(2,1). The half-life is $\tau \ln 2$ in time units, or $\ln 0.5 / \ln \phi$ in observations.

Two lessons follow. First, ϕ is not a deep structural constant: it depends on the sampling interval, and $\phi \to 1$ as $\Delta \to 0$. A slow variable observed at high frequency therefore looks like a unit root, which is one reason unit-root tests are delicate (Section 1.8). Second, $\phi = e^{-\Delta/\tau}$ is always positive. A negative ϕ, with its alternating signs, cannot come from a sampled first-order physical system and needs a genuinely discrete-time mechanism such as the cobweb of Section 1.9.

IN PLAIN WORDS. A cup of tea cools towards room temperature, and the gap shrinks

by the same fraction every minute. If the time constant is $\tau = 20$ minutes and you read the thermometer every $\Delta = 5$ minutes, then $\phi = e^{-0.25} \approx 0.78$ and the gap halves every 14 minutes. Read it every second instead and ϕ is almost 1: each reading is nearly the same as the last, though nothing about the tea has changed.

1.4 FRICTION, DAMPING AND THE ROOTS

Adding friction gives $\ddot{x} + 2\delta\dot{x} + \omega_0^2 x = 0$, whose motions are either damped oscillations (small friction) or aperiodic decay (large friction). Observe such a system every Δ time units and its free motion obeys exactly an AR(2) recursion,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2}, \qquad \phi_1 = 2e^{-\delta\Delta}\cos(\omega\Delta), \quad \phi_2 = -e^{-2\delta\Delta}, \tag{1.3}$$

with $\omega = \sqrt{\omega_0^2 - \delta^2}$. The characteristic roots of the AR(2) are $\lambda_{1,2} = \tfrac{1}{2}\big(\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}\big)$. When $\phi_1^2 + 4\phi_2 < 0$ they are complex, with modulus $r = \sqrt{-\phi_2}$ and angle $\theta = \arccos\big(\phi_1 / 2r\big)$. Then:

- the *pseudo-period* of the cycle is $2\pi/\theta$ observations;
- the autocorrelation function is a damped cosine, $\rho_k = r^k \sin(k\theta + \psi)/\sin\psi$ with $\tan\psi = \frac{1-\phi_2}{1+\phi_2}\tan\theta$;
- shocks have a damped-oscillatory impulse response, $\psi_j = r^j \sin\big((j+1)\theta\big)/\sin\theta$, which is precisely the free motion of the mechanical oscillator after a single kick.

The process is stationary iff both roots lie inside the unit circle, which for an AR(2) is the triangle $\phi_2 + \phi_1 < 1,\ \phi_2 - \phi_1 < 1,\ |\phi_2| < 1$ of Figure 1.2.

A worked case. Coefficients of roughly the size $\phi_1 = 1.34$, $\phi_2 = -0.65$ are typical of what an AR(2) fitted to annual sunspot numbers returns. Here $r = \sqrt{0.65} = 0.806$ and $\theta = 0.590$ radians, so the pseudo-period is $2\pi/\theta \approx 10.65$ years. The amplitude of the deterministic part shrinks by $r^{10.65} \approx 0.10$ per cycle (a "logarithmic decrement" of about 2.3, to use the oscillation theorists' term), and its half-life is about $\ln 0.5 / \ln r \approx 3.2$ years. A cycle that decays to a tenth of its height each period would be invisible after two or three periods; the cycle we actually observe in the data is therefore kept alive by the shocks.

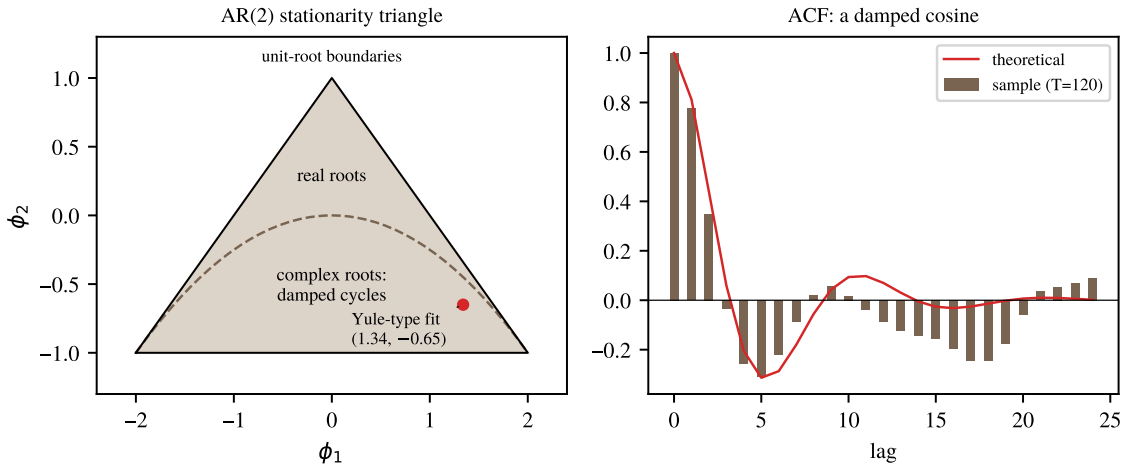


Figure 1.2: Left: the stationarity triangle of the AR(2). Below the dashed parabola $\phi_1^2 + 4\phi_2 = 0$ the roots are complex and the process cycles. Right: sample and theoretical ACF of a simulated AR(2) with $\phi_1 = 1.34,\ \phi_2 = -0.65$, a damped cosine.

1.5 THE EDGES OF THE TRIANGLE: BIFURCATIONS AND SAMUELSON'S ACCELERATOR

In the oscillation theory a *branch value* of a parameter is one at which the phase portrait changes qualitatively (Chapter VI, §5, p. 405 onwards). The book's treatment of the valve generator shows the two basic ways in which self-oscillation can appear as a parameter passes its threshold (Chapter IX, §§9–10, pp. 635–644): with *soft excitation* the oscillation grows continuously from zero amplitude, while with *hard excitation* a finite-amplitude oscillation appears abruptly and persists, with hysteresis, even when the parameter is moved back. The book predates the later name "Hopf bifurcation" (the phenomenon is often called the Andronov–Hopf bifurcation) and works in continuous time. The discrete-time translation below is therefore ours.

The stationarity region of the AR(2) is bounded by three lines (Figure 1.3), and each is a bifurcation, a place where a root reaches the unit circle:

- $\phi_1 + \phi_2 = 1$**: a root at** $+1$**.** A unit root; shocks become permanent and the process turns into a random walk (possibly with drift).
- $\phi_2 - \phi_1 = 1$**: a root at** -1**.** A period-2 oscillation (alternation from one observation to the next) becomes undamped.
- $\phi_2 = -1$ **with** $|\phi_1| < 2$**: a complex pair of modulus** 1**.** A sustained cycle of frequency $\theta = \arccos(\phi_1/2)$; this is the discrete analogue of the Hopf bifurcation, and the frictionless "centre" of Section 1.2.

The corners are double roots: $(2,-1)$ is the double unit root of an $I(2)$ process, and $(-2,-1)$ the double root -1.

A linear model has only half of the book's story. Moving a parameter across an edge changes stability, but nothing in a linear model can create a finite, self-selected amplitude: as an edge is approached the variance grows without bound, whereas in a nonlinear self-oscillator the amplitude saturates. Near the edges a stationary AR(2) also behaves like the nearby non-stationary one, which is the source of the near-unit-root problems discussed in Section 1.8.

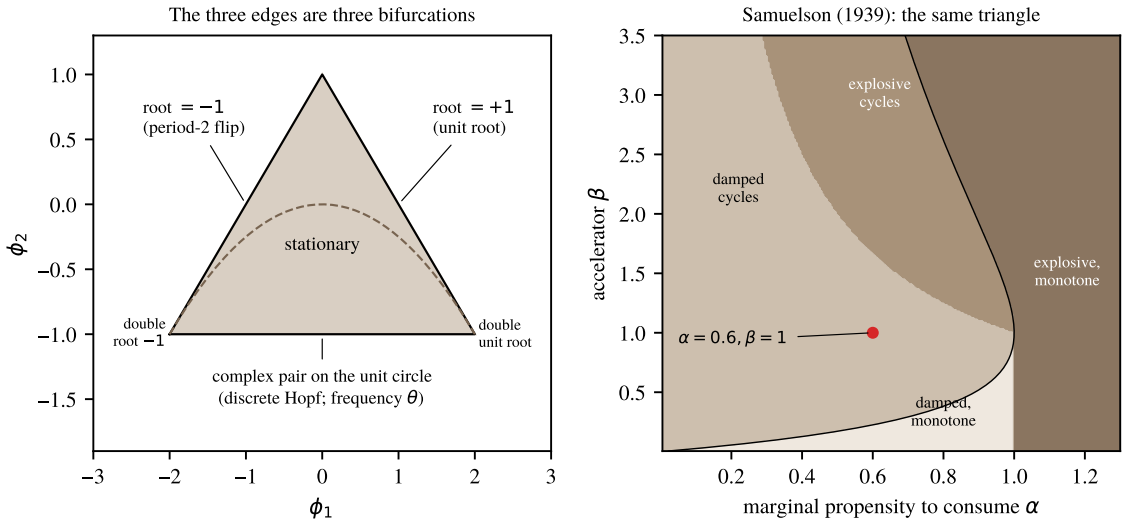


Figure 1.3: Left: the AR(2) stationarity triangle with its three edges labelled by the bifurcation that occurs there. Right: Samuelson's multiplier–accelerator model in the plane of the marginal propensity to consume α and the accelerator β; the four regions are the damped and explosive, monotone and cyclical cases.

Samuelson's accelerator as an AR(2). The classic example in economics is the multiplier–accelerator model of Samuelson (1939). Output is $Y_t = C_t + I_t + G$ with consumption $C_t = \alpha Y_{t-1}$ and induced investment $I_t = \beta(C_t - C_{t-1})$. Substituting gives

$$Y_t = \alpha(1+\beta)Y_{t-1} - \alpha\beta Y_{t-2} + G, \qquad \phi_1 = \alpha(1+\beta),\ \ \phi_2 = -\alpha\beta.$$

The three edge conditions become $\phi_1 + \phi_2 = \alpha < 1$, $\phi_2 = -1 \iff \alpha\beta = 1$, and the root at -1 is never reached for positive parameters, since $\phi_2 - \phi_1 = -\alpha(1+2\beta) < 1$ always holds. The model is therefore stationary iff $\alpha < 1$ and $\alpha\beta < 1$, and its roots are complex iff $\alpha(1+\beta)^2 < 4\beta$. These are exactly Samuelson's four regions in the right panel of Figure 1.3. For instance, $\alpha = 0.6$, $\beta = 1$ gives $\phi_1 = 1.2$, $\phi_2 = -0.6$, so $r = 0.775$ and $\theta = 0.685$ radians: damped cycles with a period of about 9.2 time units. In keeping with the discussion above, the linear model can only produce cycles that die out or explode; Hicks (1950) added a ceiling and a floor on investment, a nonlinearity, precisely to obtain bounded and persistent cycles.

1.6 ADDING THE FLUCTUATIONS: YULE'S PENDULUM

The oscillation theorists deliberately restrict themselves to *dynamic* models and neglect fluctuations. They nevertheless concede that small random perturbations are unavoidable in every real system, and that statistical models, based on the Einstein–Fokker equations, are required to study their effect (Andronov et al., 1966, p. xviii). Box–Jenkins modelling is the discrete-time statistical complement they point to. The decisive step was taken by Yule (1927): a pendulum left alone dies out, but a pendulum continually bombarded by small random blows (Yule's own image) keeps swinging with its *natural* period and an *irregular* amplitude and phase. That is the AR(2) process, and he used it to explain the apparent cycle in Wolfer's sunspot numbers. Slutsky (1937) showed independently that summing random causes manufactures cyclical behaviour from pure noise, and Frisch (1933) carried the idea into economics by separating a system's *propagation* mechanism (its damped-oscillator structure) from the random *impulses* that excite it. The same decomposition underlies the Wold representation of Chapter 3: the oscillator is the filter, the shocks are the input.

A technical footnote: a continuous-time damped oscillator driven by white noise and then sampled gives an ARMA(2,1) rather than a pure AR(2) (Bartlett, 1946; Phillips, 1959). This is one reason that mixed ARMA models are ubiquitous in applied work.

> IN PLAIN WORDS. A child on a swing: given one push, the swing's motion fades smoothly (damped oscillator). Now a crowd of children keeps giving small, random pushes: the swing never stops, always swings at roughly the same pace (its natural period), yet some swings are high and some low, and the rhythm drifts. A business-cycle economy is such a swing, and the autoregression is the equation of the swing plus the pushes.

1.7 STOCHASTIC CYCLES: SLOWLY VARYING AMPLITUDE AND PHASE

The book's main approximate method for nearly sinusoidal oscillations is due to Van der Pol (Chapter IX, §§2–3, from p. 585). Instead of solving the equation, one writes the motion as $x = a\cos t + b\sin t$, exactly as for the harmonic oscillator, but lets the coordinates a and b vary slowly with time. In the phase plane they are the coordinates of a frame rotating at the natural frequency, so that in the absence of damping and forcing every point is an equilibrium. Averaging away the rapidly oscillating terms gives *truncated equations* for the amplitude $K = \sqrt{a^2 + b^2}$ and the phase; the amplitude equation $\dot{K} = \mu\Phi(K)$ is first-order, so it can be studied on the phase line of a system with half a degree of freedom, and a stable root of $\Phi(K) = 0$ is a limit cycle. The method works when the damping is small relative to the frequency, that is, when the logarithmic decrement is small compared with unity (p. xvii).

Krylov and Bogoliubov developed the same idea independently, and Krylov and Bogoliubov (1947) made it available in English (see the survey in Chapter XI).

The time-series counterpart is the *stochastic cycle*. Consider the complex-root AR(2) of Section 1.4 with r close to 1. Define a two-component state (ψ_t, ψ_t^*) that is rotated through the angle θ, shrunk by the factor r and kicked by noise each period:

$$\psi_t = r\cos\theta\,\psi_{t-1} + r\sin\theta\,\psi_{t-1}^* + \kappa_t, \tag{1.4}$$

$$\psi_t^* = -r\sin\theta\,\psi_{t-1} + r\cos\theta\,\psi_{t-1}^* + \kappa_t^*, \tag{1.5}$$

the form used for cycles in structural time-series models (Harvey, 1989). In the rotating frame this is a pair of independent AR(1) processes, the discrete counterparts of a and b. Three consequences follow.

- *Slowly varying amplitude and phase.* For r near 1 the autocorrelation function is close to $r^k\cos(k\theta)$, the envelope r^k being the slowly varying amplitude, and the amplitude and phase wander on a time scale of order $1/(1-r)$ observations.
- *Rayleigh envelope.* For a Gaussian narrow-band process the envelope K has a Rayleigh distribution and the phase is uniform (Rice, 1944), so even a stationary cycle has no fixed amplitude, only a stable distribution of amplitudes.
- *A sharp spectral peak.* The width of the spectral peak at frequency θ is governed by $1-r$ (Figure 1.4); the sharper the pitch, the more slowly the loudness and phase drift.

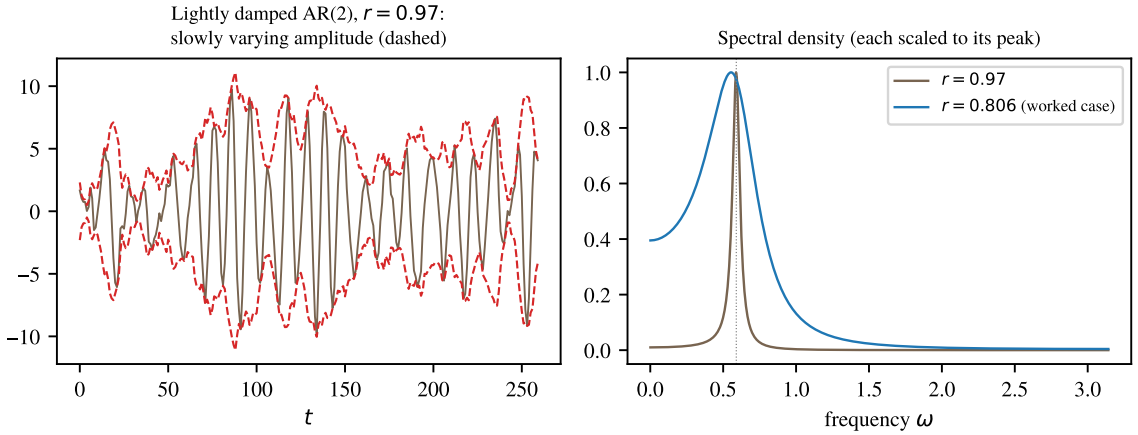


Figure 1.4: Left: a lightly damped AR(2) ($r = 0.97$, $\theta = 0.59$) with its slowly varying envelope. Right: spectral density of the AR(2) for $r = 0.97$ and for the worked case $r = 0.806$ of Section 1.4; each curve is scaled to its own peak.

A caution that the authors would insist on: the picture is only good when the damping is small. In the worked case of Section 1.4 the logarithmic decrement is about 2.3, far from small, so the

amplitude and phase decorrelate within a single cycle and the envelope description is crude. For $r = 0.97$ the decrement per cycle is about 0.32, and the picture is sound.

> IN PLAIN WORDS. A radio tuned to one station: the more lightly damped the circuit, the purer the tone it picks out, but the more slowly its loudness changes. A business cycle with r near 1 is a persistent, recognisable rhythm whose height and timing nonetheless drift unpredictably; with r well below 1 it is a faint, short-lived echo.

1.8 STABILITY, COARSENESS AND PARSIMONY

Stationary motions. The school focuses on the motions to which a system tends over long intervals: states of rest and periodic motions. A motion that lasts must be stable against small disturbances of the coordinates, and also against small changes in the equations themselves (Andronov et al., 1966, pp. xxvii–xxix). In time series the first requirement is the unit-circle condition above. The analogue of the Routh–Hurwitz tests for continuous systems (Routh, 1877; Hurwitz, 1895) is the Schur–Jury test for discrete ones (Schur, 1917; Jury, 1964); the triangle of Figure 1.2 is its AR(2) form. (The history of the continuous-time test is instructive: Hurwitz was prompted by the turbine-governor stability problem posed by Stodola, which in turn built on Vyshnegradskii's analysis of the steam-engine governor (Bissell, 2001).) A terminological warning: in the oscillation literature a *stationary motion* is a limit motion such as an equilibrium or a periodic orbit, whereas in statistics *stationarity* means invariance of moments over time. The notions are related but not identical.

Coarseness. A system is *coarse* if its qualitative behaviour survives small changes in the form of its equations; properties that vanish under such changes are, the authors argue, of no physical interest (Andronov et al., 1966, p. xxix). The formal theory of coarse systems is due to Andronov and Pontryagin (1937), and the book develops it in Chapter VI, §4 (pp. 374–405). The unit root is the canonical non-coarse case in time series: $\phi = 1$ and $\phi = 0.99$ describe qualitatively different worlds (permanent versus transitory shocks), yet no finite sample can reliably tell them apart. This is why unit-root tests have low power and why the asymptotics of near-unit-root processes need special treatment (Phillips, 1987). Invertibility of the MA part is a coarseness condition of the same kind.

Parasitic parameters. In the circuit of their Figure 8, taking account of small stray capacitance and inductance raises the order of the equation from second to fourth, yet the new terms are tiny. The authors note that no one can verify exhaustively that all such parasitic parameters leave the conclusion intact, so the final arbiter is experiment (pp. xxix–xxxi). This is the time-series problem of high-order lags with small coefficients, of near-cancelling AR and MA factors, and of the whole case for parsimony and for information criteria. The final arbiter, as for them, is out-of-sample performance.

Initial conditions. They warn that the initial conditions assigned to an idealised model must be compatible with the idealisation (pp. xx–xxi). In estimation, the analogue is the treatment of pre-sample values: conditional least squares sets the unobserved pre-sample errors to zero, an idealisation whose influence fades as the sample grows, while exact maximum likelihood handles it properly.

1.9 FROM LINEAR TO NONLINEAR MAPS: KOENIGS'S THEOREM AND THE COBWEB

Poincaré's device for studying a periodic motion is to place a short segment across the phase paths and record the successive points $s_0, s_1, s_2, \dots$ at which a path returns to it. The rule $s_{n+1} = f(s_n)$ is the *sequence function*, or point transformation; a fixed point $s^* = f(s^*)$ corresponds to a closed path, that is, a periodic motion (Andronov et al., 1966, Chapter V, §7, pp. 291–298). Sampling a continuous flow once per turn is thus the way a differential equation becomes a discrete-time model, which is the same step that turns an oscillator into an AR process. Stability of the fixed point is settled by *Koenigs's theorem* (Koenigs, 1884): s^* is stable if $|f'(s^*)| < 1$ and unstable if $|f'(s^*)| > 1$, and the test is inconclusive at $|f'(s^*)| = 1$, where higher-order derivatives decide (pp. 294–297). The proof is a one-line contraction argument: near s^* each iterate is closer to s^* than the last by a factor smaller than one.

For a linear map $y_t = c + \phi y_{t-1}$ we have $f' = \phi$, so Koenigs's theorem is the familiar condition $|\phi| < 1$ for the AR(1), and the borderline case is the unit root. The same test applies locally to a nonlinear autoregression $y_t = f(y_{t-1}) + \varepsilon_t$. One difference deserves note: for the point transformations of a planar flow the book proves that $f' > 0$ (paths cannot cross, p. 293), so convergence there is monotone, whereas for a general map f' may be negative and the iterates alternate.

Lamerey's diagram and the cobweb. Plotting f against the diagonal and following the zig-zag $s_0 \to f(s_0) \to \dots$ gives *Lamerey's diagram* (p. 294). Economists know it as the *cobweb* diagram (Ezekiel, 1938). If demand is $D_t = \alpha - \beta p_t$ and supply responds to last period's price, $S_t = \gamma + \delta p_{t-1}$, market clearing gives

$$p_t = \frac{\alpha - \gamma}{\beta} - \frac{\delta}{\beta} p_{t-1},$$

an AR(1) with $\phi = -\delta/\beta$. By Koenigs's theorem the equilibrium price is stable iff $\delta < \beta$, that is, iff supply is less price-responsive than demand (left and middle panels of Figure 1.5).

A nonlinearity changes the verdict. When the response is bounded (for example by capacity), an equilibrium with $|f'| > 1$ need not give explosive behaviour: the right panel of Figure 1.5 shows the iterates converging to a 2-cycle, which is itself a fixed point of the second iterate

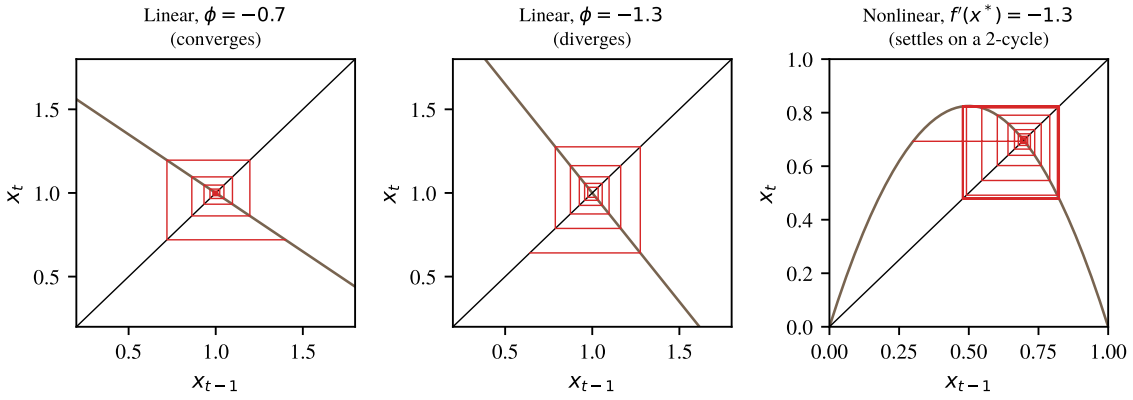


Figure 1.5: Lamerey (cobweb) diagrams. Left: a linear map with $\phi = -0.7$ converges. Middle: $\phi = -1.3$ diverges. Right: the nonlinear map $f(x) = 3.3x(1-x)$ has an unstable fixed point ($f'(x^*) = -1.3$) but, because f flattens at the extremes, the iterates settle on a stable 2-cycle between 0.48 and 0.82: a limit cycle in discrete time.

$f \circ f$, and Koenigs's theorem applies to that iterate. This is the discrete-time version of the limit cycle of Section 1.13: an oscillation whose amplitude is selected by the system and not by the initial condition. The panel is a stylised illustration and not an economic estimate.

> IN PLAIN WORDS. Farmers plant according to this year's price. A high price this year means many plantings, a glut next year and a low price; a low price means few plantings and a shortage. If farmers react mildly the swings fade, as in the left panel; if they react strongly the swings grow until some physical limit (land, storage, capacity) stops them, as in the right panel.

1.10 NEUTRAL CYCLES: VOLTERRA, LOTKA AND GOODWIN

The book uses a model from biology to show that conservativeness cannot always be decided by physical argument (Andronov et al., 1966, Chapter II, §7, pp. 142–145). The model is due to Volterra (1926) and Lotka (1925): a prey species N_1 with free growth rate ε_1 and a predator N_2 that feeds on it,

$$\dot{N}_1 = N_1(\varepsilon_1 - \gamma_1 N_2), \qquad \dot{N}_2 = -N_2(\varepsilon_2 - \gamma_2 N_1).$$

The authors find a single-valued integral, $\gamma_2 N_1 + \gamma_1 N_2 - \varepsilon_2 \ln N_1 - \varepsilon_1 \ln N_2 = \text{const}$, and an integral invariant, so the system is conservative: the paths are closed curves around the equilibrium $(\varepsilon_2/\gamma_2,\ \varepsilon_1/\gamma_1)$, and the amplitude of each cycle is fixed by the initial state (left panel of Figure 1.6).

Three lessons transfer to these notes. First, such a cycle is *neutral*: a shock moves the system onto a neighbouring orbit and it stays there, so the label of the orbit has no restoring force and

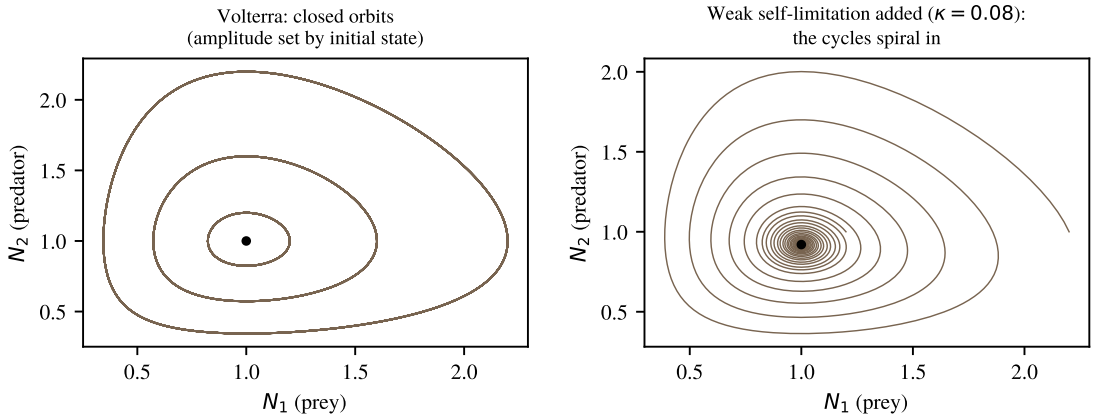


Figure 1.6: Left: the Volterra system has closed orbits whose size is set by the initial state. Right: adding a weak self-limitation term $-\kappa N_1^2$ to the prey equation destroys the closed orbits, which become a spiral: the system is not coarse.

behaves like a unit root. Second, the cycle is not coarse (Section 1.8): the right panel shows that an arbitrarily weak extra term turns the closed orbits into a spiral. Third, in economics the same equations appear in Goodwin's (1967) growth-cycle model, in which the workers' share of output plays the part of the predator and the employment rate that of the prey. A model of this kind explains why cycles exist, but its neutrality makes the amplitude of the cycle a statistical nuisance, since any estimate must contend with a quantity that never forgets its shocks.

1.11 ENERGY BALANCE: WHY A LIMIT CYCLE HAS A DEFINITE AMPLITUDE

Ginoux (2015) reports that in his 1908 lectures on wireless telegraphy Poincaré analysed the *singing arc*, a circuit whose arc has a nonlinear voltage $\theta(x')$:

$$Lx'' + \rho x' + \theta(x') + Hx = 0, \tag{1.6}$$

where $\rho x'$ is resistive damping and Hx the capacitor. Putting $y = x'$ turns this into a first-order equation in the phase plane, and Poincaré stated that sustained oscillations correspond to closed curves, stable if neighbouring paths spiral onto them (Ginoux, §1.2). To obtain a stability condition he multiplied the equation by $x'\,dt$ and integrated over one period. The inertial and capacitor terms integrate to zero, because they are conservative, leaving $\rho \oint x'^2 dt + \oint \theta(x')x'\,dt = 0$: the arc must on average supply the energy that the resistance dissipates, and the oscillation is stable iff $\oint \theta(x')x'\,dt < 0$ (§1.3). Liénard, working independently on $\ddot{x} + \omega f(x)\dot{x} + \omega^2 x = 0$, proved the existence and uniqueness of a periodic solution in 1928, and Ginoux shows that his stability condition is equivalent to Poincaré's.

Pikovsky et al. (2001, Section 2.3.1) state the same idea in the language of energy. The energy taken from a source grows with the amplitude of oscillation, and so does the energy dissipated; the steady amplitude is where the two curves intersect, and it is stable if dissipation prevails above that amplitude and supply prevails below it. Both curves pass through the origin, and they can intersect elsewhere only if the system is nonlinear. In a linear system both are proportional to the square of the amplitude, so either supply exceeds dissipation at every amplitude (growth), or falls short of it at every amplitude (decay), or the two coincide (the conservative centre).

Worked example. For the Van der Pol equation $\ddot{x}-\mu(1-x^2)\dot{x}+x=0$ with small μ, averaging over a cycle gives $\dot{K}=\frac{1}{2}\mu K\left(1-K^2/4\right)$ for the amplitude K (the amplitude equation of Section 1.7). Written as an energy balance with $E=K^2/2$, the supply is $\mu K^2/2$ and the dissipation is $\mu K^4/8$; they meet at $K=2$, whatever the initial state (Figure 1.7a,b).

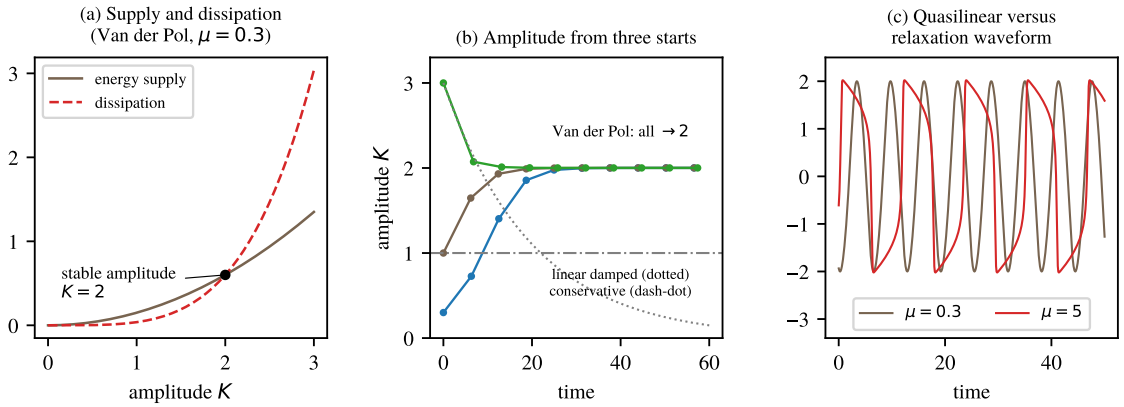


Figure 1.7: (a) Energy supply and dissipation per cycle for the Van der Pol oscillator with $\mu=0.3$; they cross at the stable amplitude $K=2$. (b) Peak amplitude from three different starts: the Van der Pol oscillator converges to $K=2$, a linear damped oscillator decays and a conservative oscillator keeps its initial amplitude. (c) Waveforms: for small μ the oscillation is nearly sinusoidal (quasilinear), for large μ it is a relaxation oscillation with a sawtooth shape and strong harmonics.

Time-series reading. For a linear AR(2) the amplitude map is $K_{t+1}=rK_t$: its only fixed point is 0 (if $r<1$), it has none (if $r>1$), or every amplitude is a fixed point ($r=1$, the centre). A nonlinear amplitude map $K_{t+1}=g(K_t)$ can instead have a nonzero fixed point with $|g'|<1$, which is a discrete-time limit cycle and is judged by Koenigs's theorem (Section 1.9). This is exactly the argument of Cartan and Cartan (1925), who proved the existence of a periodic solution of Janet's equation by plotting a first-return map, in Poincaré's manner, and showing that the curve meets the bisector at points that correspond to periodic solutions (Ginoux, 2015, §2.2.3).

1.12 SELF-SUSTAINED, FORCED OR NOISE-DRIVEN? AMPLITUDE IS STABLE, PHASE IS FREE

Pikovsky et al. (2001, Sections 2.1–2.3) define a *self-sustained oscillator* as an active system that keeps oscillating, in a rhythm fixed by its own parameters, when isolated from its environment. The practical test is to isolate it. A firefly, a plant or a human volunteer kept in constant light and temperature still shows rhythms of daily activity, so circadian rhythms are generated internally; the tides, by contrast, would vanish without the Moon, so they are *forced*. Combining this with the classes already met gives four kinds of cycle (Table 1.1).

Kind	Left alone	Amplitude and phase	Time-series counterpart
Self-sustained (limit cycle)	keeps oscillating	amplitude fixed internally; phase free	nonlinear AR with a limit cycle (threshold models; Kaldor, Goodwin-type cycles)
Forced	dies out without the force	amplitude set by the force; phase tied to it	deterministic seasonality, tied to the calendar
Noise-driven damped	dies out without shocks	Rayleigh amplitude; phase random	AR(2) cycle (Yule, Slutsky, Frisch)
Conservative	keeps oscillating	amplitude set by the initial state, never forgotten	Volterra–Lotka, Goodwin growth cycle

Table 1.1: Four kinds of cycle. The first two rows follow Pikovsky et al. (2001); the last two are developed in Sections 1.7 and the Volterra section above.

Amplitude is stable, phase is free. For a self-sustained oscillator, view the motion in a frame rotating at the natural frequency (as in Section 1.7). A disturbance of the amplitude decays, because the cycle attracts neighbouring paths; but a disturbance of the phase neither grows nor decays, because every value of the phase is equivalent (Section 2.2.2). The phase is therefore neutrally stable. Under random shocks it performs a random walk, a unit root in the phase, while the amplitude remains an ordinary stationary (AR(1)-like) deviation. The observed series is still stationary, because $\sin\varphi$ stays bounded however far the unwrapped phase wanders. By contrast, the phase of a forced system is tied to the phase of the force and is not free (Section 2.3.2).

Why this matters for identification. A noise-driven damped oscillator and a noisy limit-cycle oscillator can both have an autocorrelation function that looks like a damped cosine, and a spectrum with a peak of finite width: in the first the shocks keep the cycle alive, in the second they make the phase diffuse (the noise theory goes back to Stratonovich 1963, cited in Pikovsky et al. 2001). To second-order statistics alone the two can look alike, which is the point at which

the ACF and PACF of this document run out. Three features can discriminate between them:

- *Amplitude distribution:* concentrated near a fixed value for the limit cycle, Rayleigh (with the amplitude often near zero) for the noise-driven linear cycle.
- *Response to a one-off shock:* a limit-cycle system restores its amplitude but keeps a permanent phase shift; the linear noise-driven system's response decays to zero.
- *Waveform:* harmonics and asymmetric cycles point to a nonlinear generator (Figure 1.7c).

The isolation test is not available in macroeconomics, because an economy cannot be shielded from its shocks. This is why the debate between Frisch's *impulse* view of the business cycle and the endogenous limit-cycle models of Kaldor and Goodwin has proved so hard to settle with second-order statistics.

Seasonality is forced. By the same logic, a seasonal pattern is a forced oscillation whose phase is locked to the calendar and which would vanish if the forcing (the sun, the school year, the harvest) were removed. It is modelled deterministically (seasonal dummies) or by seasonal differencing, whereas the phase of a business cycle is not tied to any calendar. The distinction explains why the two components are treated differently in applied work.

1.13 WHAT A LINEAR MODEL CANNOT DO: LIMIT CYCLES

The same text states why a nonlinear theory is unavoidable: a linear equation cannot explain why a valve oscillator reaches a definite steady-state oscillation *independently of the initial conditions* (Andronov et al., 1966, p. xxv). In a linear system the amplitude either depends on the initial state (the centre), or decays, or explodes. A stationary ARMA is no different: the cycles it generates are sustained only by noise, with a stationary amplitude distribution and a randomly drifting phase. There is no *limit cycle*, that is, no isolated closed orbit that attracts all neighbouring paths, which was the central object that PoincarÃ© introduced and Andronov's school made the foundation of the theory of self-sustained oscillation. The text also stresses that dry (Coulomb) friction cannot be linearised even for tiny velocities, because the force jumps at zero velocity (pp. xxiii–xxiv): behaviour that changes by regime defeats any single linear model.

Pikovsky et al. (2001, Section 2.3.1) give a one-line proof of the same point: if $x(t)$ is a periodic solution of a linear system, then so is $ax(t)$ for every constant a, so a linear system that oscillates periodically is necessarily conservative, and a linear system with dissipation or a source of energy can only grow or decay. Two further remarks from that source qualify the picture. First, as the nonlinearity strengthens, a near-sinusoidal (*quasilinear*) oscillation becomes a relaxation oscillation with a strongly non-sinusoidal waveform and many harmonics

(Section 2.4 and Figure 1.7c), which a low-order linear ARMA reproduces poorly. Second, the models of the 1930s could generate only periodic self-oscillations; irregular, *chaotic* self-sustained oscillations are now known as well, so that an irregular series need not be noise-driven. Box–Jenkins modelling treats the irregularity as a random shock, which is a modelling choice and not a fact about the series.

Economics has repeatedly met this boundary. Kaldor (1940) and Goodwin (1951) built nonlinear business-cycle models whose persistent cycles are limit cycles rather than echoes of shocks, and Tong and Lim (1980) introduced threshold autoregressions in statistics explicitly to capture limit-cycle behaviour in cyclical data. The practical rule for these notes: the Box–Jenkins machinery is the right first tool, and a series that shows asymmetric cycles (fast rises and slow declines), amplitude-dependent persistence, or regime-specific behaviour (for example, near an interest-rate floor) is a signal to test for nonlinearity in the residuals before trusting a linear fit.

1.14 FROM ONE SERIES TO SEVERAL: HUYGENS'S CLOCKS AND COINTEGRATION

Pikovsky et al. (2001, Chapter 1) open their study of synchronisation with Huygens. In a letter of 26 February 1665 he described two pendulum clocks hung from a common beam that came to swing in exact opposition and, if disturbed, restored the agreement within a short time; he explained it correctly by the imperceptible motion of the beam, which couples the clocks. Rayleigh later described organ pipes of nearly equal pitch that fall into unison, or fall silent when coupled too strongly, and the study of synchronisation of triode generators by Eccles and Vincent (1920), Appleton (1922) and Van der Pol (1927) led directly to the self-oscillator theory of Andronov's school. The authors understand synchronisation as the adjustment of the rhythms of oscillating objects due to their weak interaction, and they call two oscillators $n:m$ phase-locked if $|n\varphi_1 - m\varphi_2|$ stays below a constant, a definition that allows a constant phase shift (Huygens's anti-phase) and small fluctuations of the phase difference.

There is a precise statistical analogy, which is ours and not the authors'. The phase of a self-sustained oscillator is free, that is, behaves like a unit root (Section 1.12). Two such phases wander separately, but if the oscillators are coupled, the combination $n\varphi_1 - m\varphi_2$ stays bounded. That is the definition of *cointegration* (Engle and Granger, 1987): variables that are individually integrated but share a stationary linear combination. The leash that couples the drunk and her dog (Murray, 1994) plays the part of Huygens's beam. The analogy is not an identity, since phase locking is a dynamical statement about phases and cointegration a statistical one about levels. It nevertheless suggests where Box–Jenkins modelling goes next: from a single series to systems of series (VAR and error-correction models), and it suggests reading international business-cycle comovement as a question of phase locking. The authors devote a chapter to

inferring synchronisation from data, since the interaction cannot be switched off and one can only observe the oscillators as they run freely.

1.15 A DICTIONARY BETWEEN THE TWO VOCABULARIES

Table 1.2 collects the correspondences established in this chapter. It is meant as a reference: each row pairs a concept from the oscillation literature with its counterpart in Box–Jenkins modelling.

Table 1.2: A dictionary between oscillation theory and Box–Jenkins.

Oscillation theory	Time-series counterpart
State $(x, \dot{x})$	State vector (Y_t, Y_{t-1})
Degrees of freedom = half the state variables	AR order p ($p/2$ degrees of freedom)
Equilibrium (singular point)	Unconditional mean μ
Centre (no friction)	Complex roots *on* the unit circle (non-stationary)
Stable focus (damped oscillation)	Complex roots inside the unit circle: ACF a damped cosine
Stable node (aperiodic decay)	Real roots inside the unit circle: geometric ACF
"Negative friction" (growing oscillation)	Roots outside the unit circle (explosive)
Saddle (repulsive force)	One root inside, one outside the unit circle
Routh–Hurwitz stability test	Schur–Jury test: all roots inside the unit circle
Logarithmic decrement, damping	Modulus r of the roots, half-life
Natural frequency	Angle θ of the roots, pseudo-period $2\pi/\theta$
Fluctuations (Einstein–Fokker equations)	Innovations ε_t (white noise)
Autonomous system (no explicit time)	Time-invariant coefficients
Coarse (robust) system	Roots bounded away from the unit circle; invertible MA
Parasitic parameters raising the order	Small high-order lags; near-cancelling AR/MA factors

Oscillation theory	Time-series counterpart
Initial conditions compatible with the idealisation	Pre-sample values; conditional versus exact likelihood
Check theory against experiment	Diagnostic checking, out-of-sample forecast
Qualitative integration, phase portrait	ACF/PACF as a portrait; identification
Limit cycle (amplitude independent of start)	No linear analogue; threshold autoregression
Half a degree of freedom (*RC* circuit)	AR(1), $\phi = e^{-\Delta/\tau}$ (sampled Ornstein–Uhlenbeck)
Branch value of a parameter; soft and hard excitation	Root crossing the unit circle; edges of the stationarity region
Van der Pol's slowly varying amplitude and phase	Stochastic cycle: rotating state (ψ_t, ψ_t^*), Rayleigh envelope
Sequence function, Koenigs's theorem	Nonlinear AR(1) map $y_t = f(y_{t-1})$: stable iff $\lvert f'(y^*)\rvert < 1$
Lamerey's diagram	Cobweb diagram
Conservative system, neutral cycles (Volterra)	Amplitude with no restoring force (a unit root in the orbit label)
Self-sustained oscillator (limit cycle)	Nonlinear AR with a limit cycle; not an ARMA
Forced oscillation (phase tied to the force)	Deterministic seasonality
Phase is free (neutrally stable)	Unit root in the unwrapped phase
$n:m$ phase locking	Cointegration of the phases
Energy supply equals dissipation	Amplitude fixed point $K^* = g(K^*)$ with $\lvert g'(K^*)\rvert < 1$

1.16 TWO LINEAGES THAT RARELY MET

Table 1.3 lists the two lineages side by side. The convergence of dates around 1927–1929 is striking: Yule and Slutsky in statistics, and Andronov in the physics of oscillations, were all working out how a regular cycle can be generated or sustained, one by a damped system plus noise, the other by a nonlinear system with an attracting closed orbit.

Notes on the oscillation lineage. The authors themselves record that the phase plane was first used for dynamics by Léauté, who found integral curves and limit cycles in an automatic-control problem without naming them and apparently without knowing Poincaré's work; his papers were soon almost forgotten (Andronov et al., 1966, p. 4). Andronov's own contribution, in the late 1920s, was to show that Poincaré's limit cycles describe a whole range of practical oscillatory processes, a link that Minorsky (1958) summarised as singular points $\leftrightarrow$ equilibria, limit cycles $\leftrightarrow$ stationary motions, and self-excitation $\leftrightarrow$ bifurcations (Bissell, 2001). The credit traditionally given to Andronov, which rests on a note of 1929 (Andronov, 1929), has been revised by Ginoux (2015, 2017). Ginoux reports that Poincaré had already applied his limit cycle to the singing arc in the lectures of 1908 (Poincaré, 1908), twenty years before Andronov's report to the congress of Russian physicists in August 1928, which preceded the note presented to the Paris Academy by Hadamard in October 1929. The French thread then reads as a series of near misses. Janet (1919) drew the electromechanical analogy without any reference to Poincaré, and Pomey (1920) asked Cartan to settle the existence of sustained oscillations in such a circuit; Cartan and Cartan (1925) proved it with a diagram that is exactly Poincaré's first-return map, but made no connection with his work; and Liénard, who proved existence and uniqueness of the periodic solution in 1928 (and whose stability condition Ginoux shows to be equivalent to Poincaré's), used neither the term "limit cycle" nor a reference to Poincaré, before or after 1929. Van der Pol realised that the closed curve of his 1926 paper was a limit cycle only after Andronov's note, in his Paris lectures of March 1930, where he named Poincaré as the originator of the term. The first pictures of a limit cycle on a cathode-ray oscilloscope appear in a Toulouse thesis of October 1936 (Morched-Zadeh, on a neon-lamp oscillator). Ginoux's conclusion is that the forgotten lectures produced no reaction, that France produced fundamental results but no school of nonlinear oscillations, and that Hadamard's presentation of Andronov's and Poincaré's work did little to change this. (Andronov's co-author is spelled Witt in the French notes and Vitt in the 1966 translation; they are the same person.) The lesson is of direct interest here: the same idea was found, lost and rediscovered because the links between its discoverers were not made, much as Yule, Slutsky, Walker and Wold worked in parallel on the time-series side.

The book's own history is also worth knowing. The first Russian edition of 1937 omitted the name of A. A. Vitt from its title page by what the second edition's preface calls an unfortunate mistake.[1] Lefschetz's English version of the first edition was abridged (Andronov and Khaikin, 1949); the 1966 translation used here renders the much-enlarged second edition, partly rewritten after Andronov's death in 1952 by his pupils (Andronov et al., 1966, Preface), and Bissell (2001) notes that neither English version perfectly reflects the original. Andronov insisted that researchers should know the history of their field; the present chapter is written in that spirit.

[1] The sources differ on his fate: the preface states that Vitt died in 1937, while Bissell (2001) reports that he was arrested in 1937 and died the following year in a labour camp.

The statistical lineage inside the oscillation school. The authors did not ignore noise. Their Introduction points to the Einstein–Fokker equations and, in its references, to Kolmogorov's analytical methods for random processes (Kolmogorov, 1931), to a text on statistical physics, and to a paper by Pontryagin, Andronov and Vitt, *On the statistical investigation of dynamic systems* (Andronov et al., 1966, ref. 106), listed here from the book's bibliography. In continuous time this thread developed into the theory of diffusions and of exit problems for noise-driven dynamic systems; in discrete time the same ideas reached economics through Yule, Wold and Box–Jenkins. The two threads are the same subject seen from opposite ends.

Lesson for the rest of these notes. The two lineages share their mathematics, namely linear difference and differential equations whose behaviour is governed by roots, but they differ in purpose. The physicists sought the geometry of motion and its stability, treating noise as a nuisance; the statisticians treated the noise as the object and asked how to infer the system from a single noisy record. Box–Jenkins modelling sits at the junction: it is the theory of the linear oscillator, estimated from data, driven by noise.

* * *

Year	Oscillation theory	Time series and economics
1882	Poincaré introduces the limit cycle (Poincaré, 1882)	
1892	Lyapunov: stability of motion (Lyapunov, 1892)	
1908	Poincaré's lectures on wireless telegraphy: a stable limit cycle in the singing arc (Poincaré, 1908)	
1919–20	Janet's electromechanical analogy; Pomey's textbook equation of the arc (Janet, 1919)	
1925	Cartan and Cartan: existence of a periodic solution by the first-return map (Cartan and Cartan, 1925)	
1926	Van der Pol: relaxation oscillations (Van der Pol, 1926)	
1927		Yule (damped oscillator plus shocks, sunspots) (Yule, 1927); Slutsky (Russian original)
1928	Liénard: existence and uniqueness of the periodic solution (Liénard, 1928)	
1929–30	Andronov links Poincaré's limit cycles to self-sustained oscillations (Andronov, 1929); Andronov and Vitt on its stability (Andronov and Vitt, 1930)	
1930	Van der Pol recognises his 1926 closed curve as a limit cycle (Van der Pol, 1930)	
1931	Kolmogorov: analytical methods for random processes (Kolmogorov, 1931)	Walker (Walker, 1931)
1933	First international conference on nonlinear oscillations, Paris (Van der Pol and Papaleksi); Pontryagin, Andronov and Vitt: statistical investigation of dynamic systems (Pontryagin et al., 1933)	Frisch: propagation and impulse (Frisch, 1933)
1937	*Theory of Oscillations* (Andronov, Khaikin, Vitt); Andronov and Pontryagin: coarse systems (Andronov and Pontryagin, 1937)	Slutsky in *Econometrica*
1938		Wold decomposition (Wold, 1938)
1939		Samuelson: multiplier–accelerator as an AR(2) (Samuelson, 1939)
1940		Kaldor: a nonlinear model of the trade cycle (Kaldor, 1940)
1944–45	Andronov–Maier point transformations; Moscow control seminar	Rice: envelope of narrow-band noise (Rice, 1944)
1949	Lefschetz's abridged English edition	
1950		Hicks: ceiling and floor (Hicks, 1950)
1951		Goodwin: the nonlinear accelerator (Goodwin, 1951)

CHAPTER 2

THE BIG IDEA

2.1 WHAT PROBLEM DOES BOX–JENKINS SOLVE?

Suppose you observe one variable through time: monthly inflation, daily rainfall in Pune, quarterly sales. You want to describe how it behaves and say something sensible about next month. A structural economist would build a theory (a Phillips curve, a demand function) and estimate it. Box and Jenkins (1970) proposed a different route: let the series' own history reveal its structure. The variable is treated as the output of a *stochastic process*, and the task is to find the simplest process that could have generated the data. Chapter 1 explains where this idea comes from: the physics of damped and driven oscillators.

The method is an iterative cycle of four stages:

1. **Identification**: decide whether the (stationary) series looks like an AR, MA or ARMA process, and of what order, using the ACF and PACF.
2. **Estimation**: estimate the parameters (OLS for pure AR; maximum likelihood or nonlinear least squares once MA terms appear).
3. **Diagnostic checking**: test whether the residuals behave like white noise. If not, return to stage 1.
4. **Forecasting**: use the accepted model to forecast and, crucially, evaluate forecasts out of sample.

IN PLAIN WORDS. Think of a doctor reading an ECG. She does not start from a theory of your heart; she reads the pattern in the trace, names a likely condition (identification), measures its severity (estimation), checks whether the diagnosis explains *all* the trace (diagnostics), and predicts what happens next (forecasting). If the leftover trace still shows structure, she revises the diagnosis. That loop is Box–Jenkins.

2.2 WHY ECONOMISTS TOOK IT SERIOUSLY

The method's reputation in economics rests on a striking empirical episode. Nelson (1972) showed that simple ARIMA models forecast US macroeconomic variables at least as well as the large FRB–MIT–Penn structural model, which embarrassed the econometric orthodoxy of the time. Granger and Newbold (1986) later gave the systematic account of why simple univariate benchmarks are hard to beat. The practical lesson, still valid, is that any elaborate model must beat a well-specified ARIMA benchmark before its extra complexity is justified.

* * *

CHAPTER 3

FOUNDATIONS

3.1 STOCHASTIC PROCESSES AND STATIONARITY

A time series $\{Y_t\}$ is one realisation of a stochastic process. The handwritten notes list three properties, mean, variance and covariance, and these are exactly the ingredients of *weak (covariance) stationarity*:

$$\mathbb{E}[Y_t] = \mu \quad \text{for all } t, \tag{3.1}$$

$$\operatorname{Var}(Y_t) = \gamma_0 < \infty \quad \text{for all } t, \tag{3.2}$$

$$\operatorname{Cov}(Y_t, Y_{t-k}) = \gamma_k \quad \text{depends on the lag } k \text{ only, not on } t. \tag{3.3}$$

> IN PLAIN WORDS. A stationary series is one whose "personality" does not drift. Daily temperature deviations from the seasonal norm are roughly stationary: they wander, but around a stable centre with a stable spread. A country's GDP level is not: it trends upward, so its mean today is not its mean twenty years ago.

Why does this matter? Estimating a mean or an autocorrelation from *one* observed path is only legitimate if the process looks the same through time; stationarity is what lets time-averages stand in for ensemble averages.

3.2 WHITE NOISE AND THE LAG OPERATOR

The building block is *white noise*, $\varepsilon_t \sim (0, \sigma^2)$, serially uncorrelated. Define the lag operator $LY_t = Y_{t-1}$, so that $L^k Y_t = Y_{t-k}$ and the difference operator is $\Delta = 1 - L$. Every model below is a statement about polynomials in L.

3.3 THE AUTOCORRELATION FUNCTION (ACF)

The notes define the autocorrelation at lag k as

$$\rho_k = \frac{\operatorname{Cov}(Y_t, Y_{t-k})}{\sqrt{\operatorname{Var}(Y_t)\operatorname{Var}(Y_{t-k})}} = \frac{\gamma_k}{\gamma_0}, \tag{3.4}$$

where the second equality uses stationarity ($\operatorname{Var}(Y_t) = \operatorname{Var}(Y_{t-k}) = \gamma_0$). By construction $\rho_0 = 1$ and $|\rho_k| \le 1$. The sample version is $\hat{\rho}_k = \sum_{t=k+1}^{T}(Y_t - \bar{Y})(Y_{t-k} - \bar{Y}) \big/ \sum_{t=1}^{T}(Y_t - \bar{Y})^2$. Bartlett (1946) derived its sampling variability: for a white-noise series $\hat{\rho}_k$ is approximately $N(0, 1/T)$, which is the origin of the familiar $\pm 1.96/\sqrt{T}$ confidence bands in every ACF plot.

3.4 THE PARTIAL AUTOCORRELATION FUNCTION (PACF)

The ACF at lag k mixes direct and indirect influence: Y_t is correlated with Y_{t-2} partly because both are correlated with Y_{t-1}. The PACF strips the indirect channel out. In the words of your notes, it is the OLS estimate on Y_{t-k}; precisely, ϕ_{kk} is the *last* coefficient in the regression

$$Y_t = c + \phi_{k1} Y_{t-1} + \phi_{k2} Y_{t-2} + \cdots + \phi_{kk} Y_{t-k} + u_t. \tag{3.5}$$

Equivalently, it solves the Yule–Walker equations, computed efficiently by the Levinson–Durbin recursion (Levinson, 1947; Durbin, 1960). The idea of testing successive partial correlations to find an autoregressive order goes back to Quenouille (1949).

> IN PLAIN WORDS. You and your grandfather share a surname trait. Your correlation with him is high mainly *because* he passed it to your father, who passed it to you. The *partial* correlation with your grandfather, once your father is accounted for, measures whether he has any *direct* influence left. The PACF asks exactly this of each lag in turn.

3.5 THE WOLD DECOMPOSITION: WHY ARMA AT ALL?

The theoretical licence for the whole enterprise is Wold's (1938) theorem: any covariance-stationary, purely nondeterministic process can be written as an infinite moving average of white noise,

$$Y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \qquad \sum_j \psi_j^2 < \infty, \ \psi_0 = 1. \tag{3.6}$$

This infinite sequence of ψ_j cannot be estimated from finite data. ARMA models approximate the lag polynomial $\psi(L)$ by a ratio of two short polynomials, $\psi(L) \approx \theta(L)/\phi(L)$. This is why ARMA is called *parsimonious*: a handful of parameters stands in for an infinite sequence. The idea that random shocks, once summed or averaged, produce apparently regular cycles is due to

Slutsky (1937) (Russian original 1927) and Yule (1927), the two papers from which the modern subject descends.

* * *

CHAPTER 4

THE MODEL FAMILY

4.1 AUTOREGRESSIVE MODELS, AR(p)

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t, \qquad \phi(L) Y_t = c + \varepsilon_t. \tag{4.1}$$

The process is stationary iff all roots of $\phi(z) = 0$ lie outside the unit circle. For AR(1): $\mu = c/(1-\phi)$, $\gamma_0 = \sigma^2/(1-\phi^2)$ and

$$\rho_k = \phi^k,$$

so the ACF *decays geometrically* (with alternating signs if $\phi < 0$). The PACF has $\phi_{11} = \phi$ and $\phi_{kk} = 0$ for $k > 1$: it *cuts off* after lag p.

> IN PLAIN WORDS. Your bank balance of unspent monthly budget: this month's leftover depends on last month's leftover, a fixed share ϕ of it carries over, plus fresh income and expenses. Or: persistent inflation in India. If inflation was high last quarter, it is likely high this quarter; a shock fades gradually, like an echo in a hall, but never abruptly. The echo's persistence is ϕ: with $\phi = 0.9$ the echo lasts long, with $\phi = 0.2$ it vanishes quickly.

4.2 MOVING-AVERAGE MODELS, MA(q)

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}. \tag{4.2}$$

This is always stationary. For MA(1), $\rho_1 = \theta/(1+\theta^2)$ (whose absolute value cannot exceed 0.5) and $\rho_k = 0$ for $k > 1$. The ACF therefore *cuts off* after lag q, while the PACF *decays gradually*. Uniqueness requires *invertibility* ($|\theta| < 1$ for MA(1)): without it, θ and $1/\theta$ give identical autocorrelations.

> IN PLAIN WORDS. A one-day mega-sale or a strike is a *shock* that touches today's sales and tomorrow's (customers who postponed purchases), and then is gone. Its memory has a hard end: it affects exactly q periods. An AR process has a memory that fades forever; an MA process has a memory with a deadline.

4.3 ARMA(p,q) AND ARIMA(p,d,q)

$$\phi(L)Y_t = c + \theta(L)\varepsilon_t, \qquad \phi(L)(1-L)^d Y_t = c + \theta(L)\varepsilon_t \text{ (ARIMA)}. \tag{4.3}$$

For ARMA(1,1), $\rho_1 = \dfrac{(1+\phi\theta)(\phi+\theta)}{1+2\phi\theta+\theta^2}$ and $\rho_k = \phi\rho_{k-1}$ for $k \geq 2$. Both functions decay, so neither cuts off cleanly, and this is the "mixed pattern" of your notes. The "I" in ARIMA stands for *integrated*: the series must be differenced d times to become stationary (Box and Jenkins, 1970).

> IN PLAIN WORDS. A drunk walking home: each step is the previous position plus a random stumble. Her position is non-stationary (it wanders anywhere), but her *steps* are stationary noise. Differencing converts positions into steps. Murray (1994) extends this to "a drunk and her dog" to explain cointegration, which is the natural next topic after univariate modelling.

4.4 THE IDENTIFICATION TABLE

Process	ACF	PACF
AR(p)	tails off (geometric / damped waves)	**cuts off after lag** p
MA(q)	**cuts off after lag** q	tails off
ARMA(p,q)	tails off	tails off
Non-stationary (unit root)	very slow, near-linear decay from ≈ 1	large spike at lag 1, then small

Table 4.1: Theoretical signatures. The decisive distinction is *cut-off* versus *tail-off*, not "exponential" versus "not exponential."

CORRECTIONS TO THE HANDWRITTEN NOTES

The handwritten page on MA identification states that the PACF decays exponentially and the ACF dies down non-exponentially, and the AR page states the reverse. These are swapped relative to the standard result in Table 4.4: an **AR** process has a decaying ACF and a cut-off PACF; an **MA** process has a cut-off ACF and a decaying PACF. Note also that an ARMA order such as $(4,3)$ cannot be read off the plots, since both functions tail off; this is why information criteria (Chapter 7) are used to settle the orders.

* * *

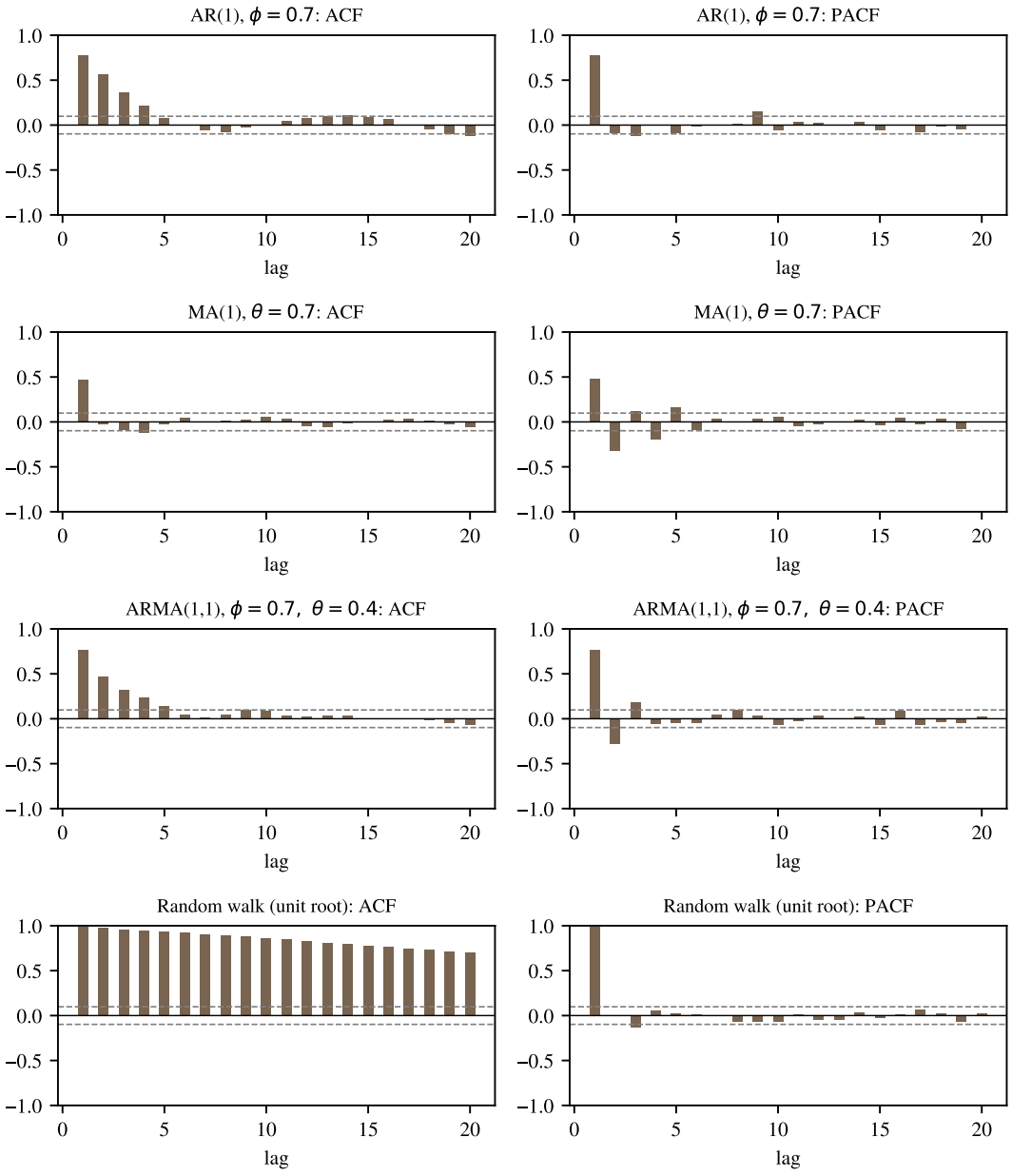


Figure 4.1: Sample ACF and PACF of simulated series ($T = 400$, seed fixed). Dashed lines are the $\pm 1.96/\sqrt{T}$ Bartlett bands. The AR(1) PACF cuts off at lag 1; the MA(1) ACF cuts off at lag 1; the ARMA(1,1) shows both tailing off; the random walk has an ACF close to one that dies very slowly.

Chapter 5

Stage 1: Identification in Practice

5.1 Step zero: is the series stationary?

Before reading the ACF and PACF for AR or MA orders, one must settle stationarity. A slowly decaying ACF near 1 (as in your ARIMA sketch) is the classic symptom of a unit root. Formal tests:

- **Augmented Dickey–Fuller (ADF)** (Dickey and Fuller, 1979): null of a unit root.
- **Phillips–Perron** (Phillips and Perron, 1988): same null, with a non-parametric correction for serial correlation.
- **KPSS** (Kwiatkowski et al., 1992): reversed null of stationarity, useful as a confirmation.

In our simulated random walk the ADF statistic is 0.05 ($p = 0.96$): the unit root is not rejected. After first differencing, the statistic is -12.91, a decisive rejection.

5.2 Why it matters: spurious regression and trend versus difference stationarity

Regressing one trending series on another produces high R^2 and significant t-statistics even when the series are unrelated (Granger and Newbold, 1974). Nelson and Plosser (1982) found that most US macroeconomic series are better described as difference-stationary (stochastic trend) than trend-stationary, which is why differencing, rather than detrending, became the default.

Do not over-difference. Differencing a series that is already stationary introduces a unit root in the MA part (an MA coefficient near -1), inflates the variance and produces a non-invertible model. Difference only as many times as the tests and the plots demand, usually $d \leq 2$.

* * *

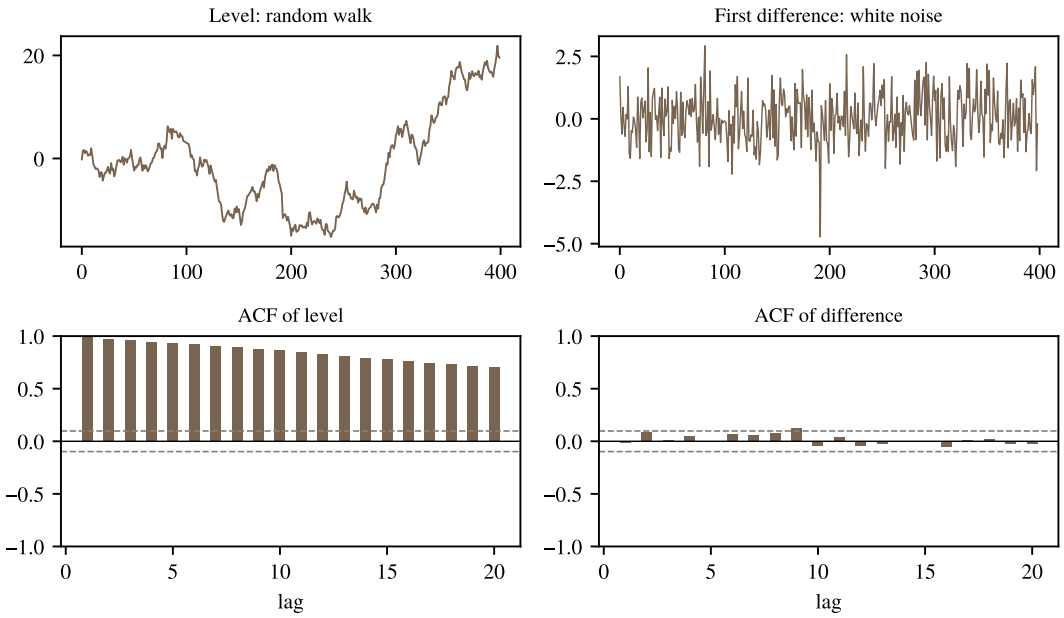


Figure 5.1: Differencing a random walk. The level has an ACF stuck near one; its first difference is white noise.

Chapter 6

Stage 2: Estimation

6.1 Autoregressions: OLS and Yule–Walker

Your notes consider candidate models with one, two and three lags, e.g. $y_t = \alpha_1 + \alpha_2 y_{t-1} + \varepsilon_t$, up to the AR(3) $y_t = \alpha_0 + \sum_{k=1}^{3} \alpha_k y_{t-k} + \varepsilon_t$. (A notational point: the sum should run to $k = 3$, carry coefficients indexed by k, and include ε_t.) Each is a linear regression of y_t on its own lags. OLS is consistent and asymptotically normal for stationary autoregressions with white-noise errors (Mann and Wald, 1943), but it is *biased in small samples* because regressors are lagged dependent variables. The older Yule–Walker method (Yule, 1927; Walker, 1931) equates sample and theoretical autocovariances and gives asymptotically equivalent estimates.

Because the regressors are lagged outcomes, the Durbin–Watson statistic is invalid for checking residual autocorrelation here, and one uses Durbin's h (Durbin, 1970) or, better, the Breusch–Godfrey test (Chapter 8).

6.2 Models with MA terms: maximum likelihood

OLS fails for MA terms because the lagged errors $\varepsilon_{t-1}, \dots$ are unobserved. Estimation therefore uses *conditional* least squares (setting pre-sample errors to zero) or *exact maximum likelihood*, computed by writing the model in state-space form and running the Kalman filter (Kalman, 1960; Ansley, 1979; Gardner et al., 1980). A simple consistent starting-value method for ARMA models is the regression procedure of Hannan and Rissanen (1982). With $T = 400$ simulated observations from an ARMA(1,1) with true $\phi = 0.7$, $\theta = 0.4$, maximum likelihood gives $\hat{\phi} = 0.62$ and $\hat{\theta} = 0.44$.

* * *

Chapter 7

The Role of Information Criteria

Plots suggest candidates; criteria adjudicate. All have the form "goodness of fit minus a penalty for complexity":

$$\mathrm{AIC} = -2\ln \hat{L} + 2k, \tag{7.1}$$

$$\mathrm{SIC\ (BIC)} = -2\ln \hat{L} + k\ln T, \tag{7.2}$$

$$\mathrm{HQIC} = -2\ln \hat{L} + 2k\ln\ln T, \tag{7.3}$$

$$\mathrm{FPE} = \hat{\sigma}^2 \frac{T+k}{T-k}, \tag{7.4}$$

where k is the number of parameters. The **lowest** value wins, so the rule in your notes is right. The origins: FPE and AIC are due to Akaike (1969, 1970, 1974), and the latter is derived from the Kullback–Leibler divergence (Kullback and Leibler, 1951); SIC is the Bayesian criterion of Schwarz (1978); HQIC is due to Hannan and Quinn (1979).

> In plain words. Fitting a model is like buying a suit. Tailoring it very tightly to today's body (a model with many parameters) looks perfect now but splits when you move. The penalty term is the tailor's warning: "more stitches are only worth it if they fit better."

7.1 Which criterion, when? Correcting "annual = AIC, high frequency = SIC"

The shorthand in the notes is really about *sample size*. SIC penalises complexity more heavily ($\ln T > 2$ once $T \geq 8$), so it is *consistent*: with enough data it selects the true order if the true model is in the candidate set (Hannan, 1980). AIC is *asymptotically efficient* for prediction (Shibata, 1980) but tends to overfit (Shibata, 1976). High-frequency data give large T, where consistency matters, so SIC is attractive; annual data give small T, where AIC's lighter penalty is safer. For very small samples use the corrected AIC_c (Sugiura, 1978; Hurvich and Tsai, 1989). In practice report both. If they agree, you can be confident. Burnham and Anderson (2002) give a full treatment.

Two rules for valid comparison. First, all models must be estimated on the *same* sample, so trim initial observations to a common start when lag lengths differ. Second, criteria cannot compare models with different orders of differencing, because the dependent variable differs.

Model	AIC	SIC	HQIC
ARMA(0,0)	1445.3	1453.2	1448.4
ARMA(0,1)	1162.1	1173.9	1166.8
ARMA(1,0)	1099.8	1111.6	1104.5
ARMA(1,1)	**1059.5**	**1075.3**	**1065.8**
ARMA(1,2)	1061.3	1081.0	1069.1
ARMA(2,1)	1061.3	1081.1	1069.2
ARMA(2,2)	1062.9	1086.6	1072.3
ARMA(3,3)	1061.6	1093.1	1074.1

Table 7.1: Information criteria for the simulated series (true model ARMA(1,1)); minima in bold. All three criteria pick the true order; the over-fitted models pay a penalty for parameters that add almost no likelihood.

* * *

CHAPTER 8

STAGE 3: DIAGNOSTIC CHECKING

The central idea: if the model has captured all the structure, the residuals $\hat{\varepsilon}_t$ should be indistinguishable from white noise. Your notes state this as "the model should satisfy its regression assumptions" and list homoscedasticity, leaving the second item blank. The complete list is below.

1. **No serial correlation in residuals** (the missing item 2). Inspect the residual ACF, and test with the Ljung–Box statistic (Ljung and Box, 1978), an improvement on the Box–Pierce statistic (Box and Pierce, 1970):

$$Q(m) = T(T+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{T-k} \overset{a}{\sim} \chi^2_{m-p-q}.$$

   The degrees of freedom are reduced by the number of estimated ARMA parameters. The Breusch–Godfrey LM test (Breusch, 1978; Godfrey, 1978) is the regression-based alternative and remains valid with lagged dependent variables.

2. **Constant variance.** Test for ARCH effects in the residuals with Engle's LM test (Engle, 1982) or the McLeod–Li test on squared residuals (McLeod and Li, 1983); White's test (White, 1980) is the general option. Volatility clustering, common in financial and inflation data, signals that a GARCH-type model is needed alongside the ARMA mean.

3. **Normality.** Jarque–Bera (Jarque and Bera, 1980). Normality is needed for exact intervals and likelihood-based inference, not for consistency of the point estimates.

4. **Stability.** All AR roots outside the unit circle and MA roots outside the unit circle (invertibility); CUSUM tests for parameter stability (Brown et al., 1975).

5. **Parsimony.** Check for *common factors*: if an AR root and an MA root nearly cancel, the model is over-parameterised, and one of each should be dropped. Also try adding one more lag and confirm it is insignificant.

A clarification on the note "we drop multicollinearity." Lagged values of a series are naturally

correlated with each other, so multicollinearity among lags is not an assumption to be checked. The relevant danger is *redundancy* (the common-factor problem above), not collinearity in the regression sense.

For the simulated ARMA(1,1) of Chapters 6 and 7, the Ljung–Box test at $m = 10$ gives $Q = 7.5$ with $p = 0.48$: no evidence of leftover autocorrelation, so the model passes.

> IN PLAIN WORDS. Diagnostics are the "leftover test." After cooking, if you can still taste an unbalanced flavour in the dish, you missed something. If the leftover residuals still carry a pattern, the model has not extracted everything and must be revised.

* * *

CHAPTER 9

STAGE 4: FORECASTING POWER

9.1 HOW FORECASTS ARE BUILT

The optimal (minimum mean-squared-error) linear forecast is the conditional expectation given the past (Kolmogorov, 1941; Wiener, 1949; Whittle, 1963). For a stationary AR(1),

$$\hat{Y}_{T+h|T} = \mu + \phi^h (Y_T - \mu), \qquad \operatorname{Var}(e_{T+h|T}) = \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2}. \tag{9.1}$$

Two features follow. The forecast *reverts to the mean* as h grows, and the forecast-error variance rises toward the unconditional variance γ_0. For an MA(q), forecasts beyond horizon q equal the mean exactly. For an ARIMA with $d = 1$ there is no mean reversion; forecast intervals widen without bound.

> IN PLAIN WORDS. Predicting tomorrow's temperature from today's is accurate. Predicting the temperature thirty days ahead from today's reading is hardly better than quoting the monthly average: the memory has faded. That is mean reversion, and every honest forecast reveals it through widening uncertainty bands.

9.2 EVALUATING FORECASTS: IN-SAMPLE FIT IS NOT ENOUGH

A model can fit the past beautifully and still forecast poorly. Hold out the last observations, estimate on the rest and compare forecasts with outcomes using

$$\text{RMSE} = \sqrt{\tfrac{1}{H}\textstyle\sum e_h^2}, \quad \text{MAE} = \tfrac{1}{H}\textstyle\sum |e_h|, \quad \text{MASE (Hyndman and Koehler, 2006)},$$

and Theil's U (Theil, 1966). To test whether the *difference* in accuracy between two models is statistically significant, use the Diebold–Mariano test (Diebold and Mariano, 1995). Prefer rolling-origin evaluation to a single split (Tashman, 2000).

On the 20 hold-out observations, the ARMA(1,1) obtains RMSE = 1.96 and MAE = 1.63,

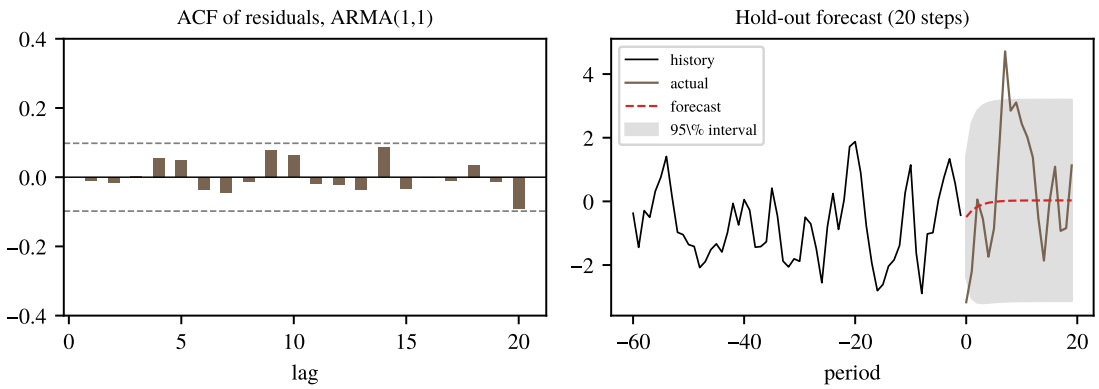


Figure 9.1: Left: residual ACF of the fitted ARMA(1,1), with all spikes inside the bands. Right: 20-step hold-out forecast with 95% interval.

against RMSE $= 2.02$ for the naive forecast of the historical mean. The modest gain reflects the theory above: with $\phi = 0.7$ the memory fades within a few steps, so only the first few forecasts can beat the mean by much.

ON THE FINAL HANDWRITTEN PAGE

The last page is faint. I read its two points as: forecasting verifies that the model captures the variable's dynamics, and the exercise is the first step toward dynamic modelling. Both are sound. Forecast performance is the ultimate arbiter because a model that passes every in-sample diagnostic can still fail on new data.

* * *

CHAPTER 10

PUTTING IT TOGETHER

10.1 PRACTICAL CHECKLIST

1. Plot the series. Look for trend, seasonality, outliers, breaks and changing variance. Consider logs for multiplicative growth.
2. Test for unit roots (ADF with KPSS as a check); difference if needed; do not over-difference.
3. Inspect ACF and PACF; list two or three candidate orders.
4. Estimate the candidates on a common sample; compare AIC and SIC.
5. Diagnose the leading candidate: Ljung–Box, ARCH test, normality, roots, an extra-lag test.
6. Forecast out of sample; compare with a naive benchmark and with a rival model.
7. If any step fails, return to step 3. The cycle is the method.

10.2 SOFTWARE

```
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.stats.diagnostic import acorr_ljungbox

adfuller(y) # unit-root test
fit = ARIMA(y, order=(1, 0, 1), trend="c").fit()
print(fit.summary()) # estimates, AIC, BIC
acorr_ljungbox(fit.resid, lags=[10], model_df=2)
fc = fit.get_forecast(20) # forecasts + intervals
```

10.3 LIMITS OF THE METHOD

Box–Jenkins is univariate and linear. It cannot explain *why* a variable moves, handles seasonality only through seasonal extensions (SARIMA), and assumes a constant structure. Extensions include ARCH/GARCH for volatility, VAR for several variables (Sims, 1980), cointegration for related non-stationary series, and state-space models (Harvey, 1989).

* * *

Annotated Reading Path

Conceptual foundation. Pikovsky et al. (2001) for self-sustained oscillators, phase and synchronisation (Chapters 1–2); Ginoux (2015) for Poincaré's forgotten lectures and the French reception; Andronov et al. (1966) for the theory of oscillations (Chapters I, IV, V §7, VI §§4–5 and IX are the most relevant); Krylov and Bogoliubov (1947) for averaging; Bissell (2001) for Andronov and his school; Ginoux (2017) for the earlier French history; Yule (1927) and Frisch (1933) for the bridge to stochastic models.

Origins (1920s–1940s). Yule (1927), Slutsky (1937), Walker (1931), Wold (1938).

Synthesis (1970). Box and Jenkins (1970); modern edition: Box et al. (2015).

Textbook theory. Brockwell and Davis (1991), Hamilton (1994); for applied economics, Enders (2014).

Practical forecasting. Hyndman and Athanasopoulos (2021).

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